12.7 Modeling with Surface Area and Volume

Name _____ Period ____ Date _____

Geometry 12.7 (30-minute plan) · Student sheet

I can…

- find and use density.
- compare the surface areas and volumes of similar solids.
- find the box with the least surface area for a given volume.

Vocabulary · fill in the missing word

Density — the _____ of matter that an object has in a given unit of volume: mass ÷ volume

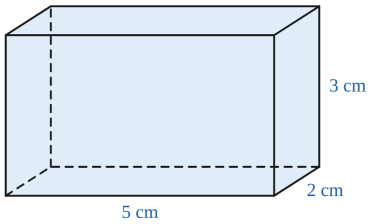
Similar solids — two solids of the same type with equal ratios of _____ linear measures

Part 1 · Notes and examples (fill in each note, then work the example)

Density = _____ ÷ _____

Example 1

A metal block is 2 cm by 3 cm by 5 cm and has a mass of 237 grams. Find its density.



Similar solids: with scale factor k, the surface areas are in the ratio _____ and the volumes are in the ratio _____.

Example 2

Two similar prisms have a scale factor of 3 : 5. The smaller has a surface area of 90 ft² and a volume of 54 ft³. Find the surface area and volume of the larger.

Least surface area: for a given volume, the box with the least surface area is a _____.

Example 3

Which box with a volume of 64 in.³ has the least surface area: 4 by 4 by 4, 2 by 4 by 8, or 1 by 8 by 8?

Part 2 · Partner practice

Partner practice

Two similar cones have a scale factor of 1 : 4. The smaller has a volume of 12 cm³. Find the volume of the larger.